theorem concerning Θ. A complete classification of these measures without entropy hypotheses would imply the full conjecture of Littlewood.

In this volume we will develop the minimal background needed for the ergodic approach to continued fractions (see Chap. 3) as well as the basic theorems concerning the action of the diagonal subgroup A on the quotient space $\mathrm{SL}_2(\mathbb{Z})\backslash \mathrm{SL}_2(\mathbb{R})$ (see Chap. 9). We will also describe the connection between these two topics, which will help us to prove results about the continued fraction expansion and about the action of A.

1.6 Integral Quadratic Forms

An important topic in number theory, both classical and modern, is that of integral quadratic forms. A quadratic form $Q(x_1, \dots, x_n)$ is said to be *integral* if its coefficients are integers.

A natural problem[(4)] is to describe the range $Q(\mathbb{Z}^n)$ of an integral quadratic form evaluated on the integers. A classical theorem of Lagrange[(5)] on the sum of four squares says that $Q_0(\mathbb{Z}^4) = \mathbb{N}_0$ if

$$Q_0(x_1, x_2, x_3, x_4) = x_1^2 + x_2^2 + x_3^2 + x_4^2,$$

solving the problem for a particular form.

More generally, Kloosterman, in his dissertation of 1924, found an asymptotic formula for the number of expressions for an integer in terms of a positive definite quadratic form Q in five or more variables and deduced that any large integer lies in $Q(\mathbb{Z}^n)$ if it satisfies certain congruence conditions. The case of four variables is much deeper, and required him to make new deep developments in analytic number theory; special cases appeared in [201] and the full solution in [202], where he proved that an integral definite quadratic form Q in four variables represents all large enough integers a for which there is no *congruence obstruction*. Here we say that $a \in \mathbb{N}$ has a congruence obstruction for the quadratic form $Q(x_1, \dots, x_n)$ if a modulo d is not a value of $Q(x_1, \dots, x_n)$ modulo d for some $d \in \mathbb{N}$.

The methods that are usually applied to prove these theorems are purely number-theoretic. Ellenberg and Venkatesh [83] have introduced a method that combines number theory, algebraic group theory, and ergodic theory to prove results in this field, leading to a different proof of the following special case of Kloosterman's Theorem.

Theorem (Kloosterman). Let Q be a positive definite quadratic form with integer coefficients in at least 6 variables. Then all large enough integers that do not fail the congruence conditions can be represented by the form Q.

That is, if $a \in \mathbb{N}$ is larger than some constant that depends on Q and for every $d > 0$ there exists some $x_d \in \mathbb{Z}^n$ with $Q(x_d) = a$ modulo d, then there